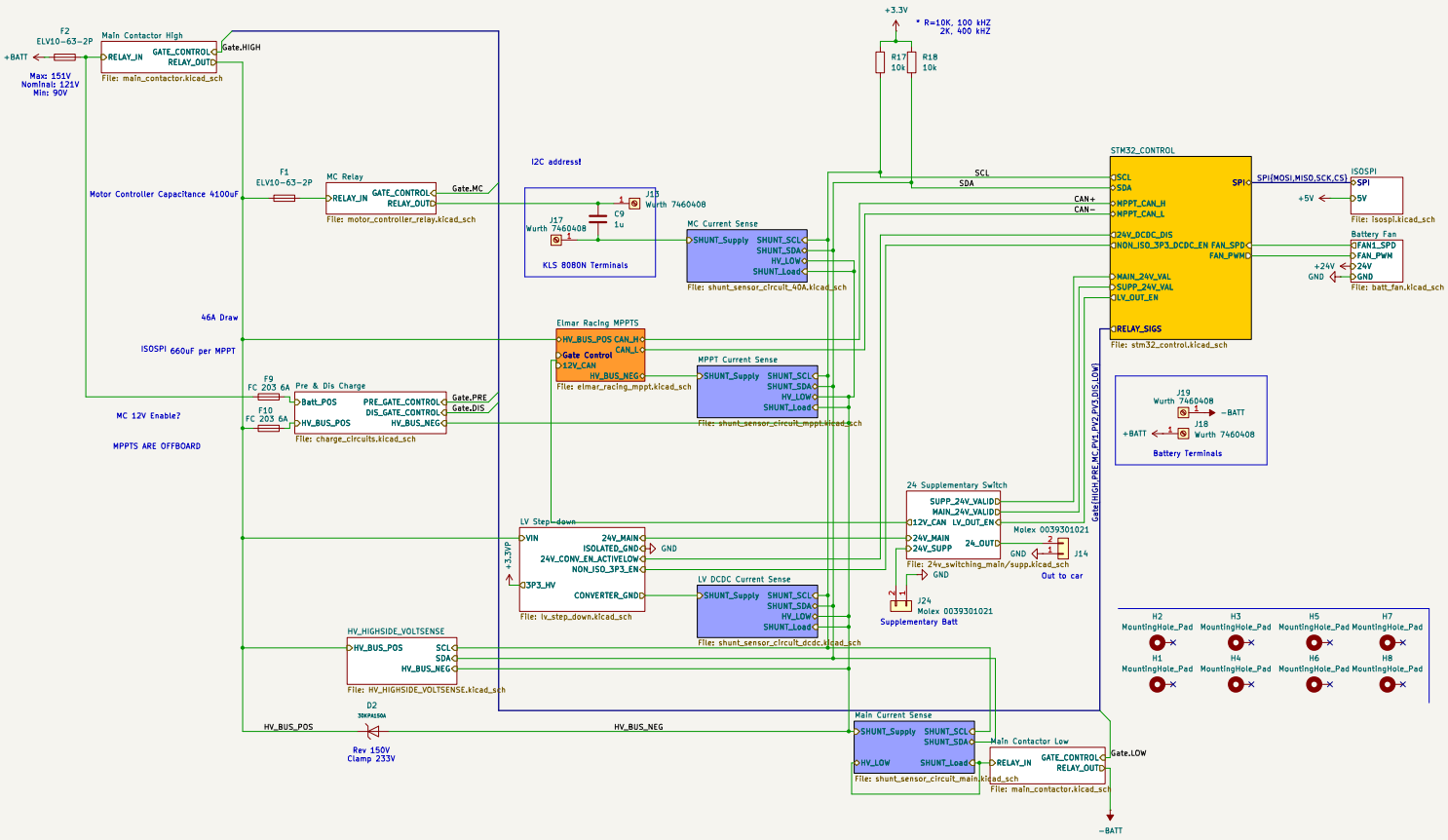
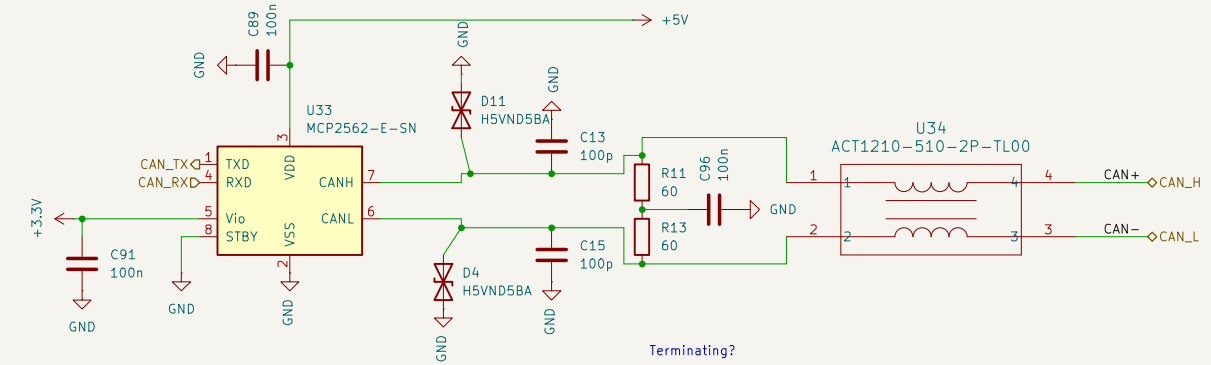
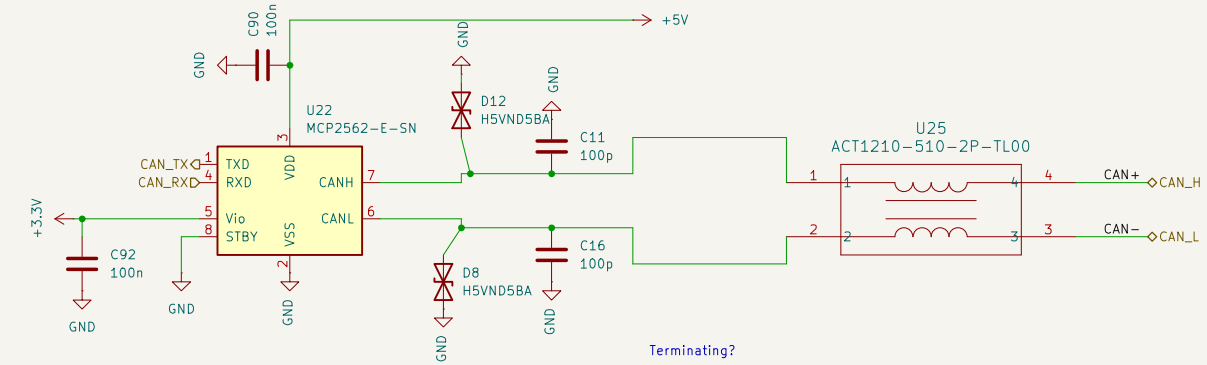

F2
ELV10-63-2P
Main Contactor High
+BATT
RELAY_IN
GATE_CONTROL
RELAY_OUT
Gate.HIGH
File: main_contactor.kicad_sch
Max: 151V
Nominal: 121V
Min: 90V
Motor Controller Capacitance 4100uF
F1
ELV10-63-2P
MC Relay
RELAY_IN
GATE_CONTROL
RELAY_OUT
Gate.MC
File: motor_controller_relay.kicad_sch
I2C address!
J17
Wurth 7460408
C9
1u
J16
Wurth 7460408
KLS 8080N Terminals
MC Current Sense
SHUNT_Supply
SHUNT_SCL
SHUNT_SDA
HV_LOW
SHUNT_Load
File: shunt_sensor_circuit_40A.kicad_sch
+3.3V
* R=10K, 100 kHZ
2K, 400 kHZ
R17
10k
R18
10k
SCL
SDA
CAN+
CAN-
STM32_CONTROL
SCL
SDA
MPPT_CAN_H
MPPT_CAN_L
24V_DCDC_DIS
NON_ISO_3P3_DCDC_EN
MAIN_24V_VAL
SUPP_24V_VAL
LV_OUT_EN
RELAY_SIGS
SPI
FAN_SPD
FAN_PWM
SPI(MOSI,MISO,SCK,CS)
ISOSPI
SPI
+5V
5V
File: isospi.kicad_sch
Battery Fan
FAN1_SPD
FAN_PWM
24V
GND
+24V
GND
File: batt_fan.kicad_sch
File: stm32_control.kicad_sch
46A Draw
ISOSPI 660uF per MPPT
MC 12V Enable?
MPPTS ARE OFFBOARD
Elmar Racing MPPTS
HV_BUS_POS
CAN_H
Gate Control
CAN_L
12V_CAN
HV_BUS_NEG
File: elmar_racing_mppt.kicad_sch
MPPT Current Sense
SHUNT_Supply
SHUNT_SCL
SHUNT_SDA
HV_LOW
SHUNT_Load
File: shunt_sensor_circuit_mppt.kicad_sch
F9
FC 203 6A
F10
FC 203 6A
Pre & Dis Charge
Batt_POS
HV_BUS_POS
PRE_GATE_CONTROL
DIS_GATE_CONTROL
HV_BUS_NEG
Gate.PRE
Gate.DIS
File: charge_circuits.kicad_sch
J19
Wurth 7460408
-BATT
J18
+BATT
Wurth 7460408
Battery Terminals
24 Supplementary Switch
SUPP_24V_VALID
MAIN_24V_VALID
12V_CAN
LV_OUT_EN
24V_MAIN
24V_SUPP
24_OUT
GND
Molex 0039301021
J14
Out to car
File: 24v_switching_main/supp.kicad_sch
Gate(HIGH,PRE,MC,PV1,PV2,PV3,DIS,LOW)
LV Step-down
VIN
24V_MAIN
ISOLATED_GND
24V_CONV_EN_ACTIVELOW
NON_ISO_3P3_EN
3P3_HV
CONVERTER_GND
GND
+3.3VP
File: lv_step_down.kicad_sch
LV DCDC Current Sense
SHUNT_Supply
SHUNT_SCL
SHUNT_SDA
HV_LOW
SHUNT_Load
File: shunt_sensor_circuit_dcdc.kicad_sch
J24
Molex 0039301021
Supplementary Batt
HV_HIGHSIDE_VOLTSENSE
HV_BUS_POS
HV_BUS_NEG
SCL
SDA
File: HV_HIGHSIDE_VOLTSENSE.kicad_sch
D2
30KPA150A
Rev 150V
Clamp 233V
HV_BUS_POS
HV_BUS_NEG
Main Current Sense
SHUNT_Supply
SHUNT_SCL
SHUNT_SDA
HV_LOW
SHUNT_Load
File: shunt_sensor_circuit_main.kicad_sch
Main Contactor Low
RELAY_IN
GATE_CONTROL
RELAY_OUT
Gate.LOW
File: main_contactor.kicad_sch
-BATT
H2 MountingHole_Pad
H3 MountingHole_Pad
H5 MountingHole_Pad
H7 MountingHole_Pad
H1 MountingHole_Pad
H4 MountingHole_Pad
H6 MountingHole_Pad
H8 MountingHole_Pad
Sheet: /
File: BatteryBoard.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 1/27

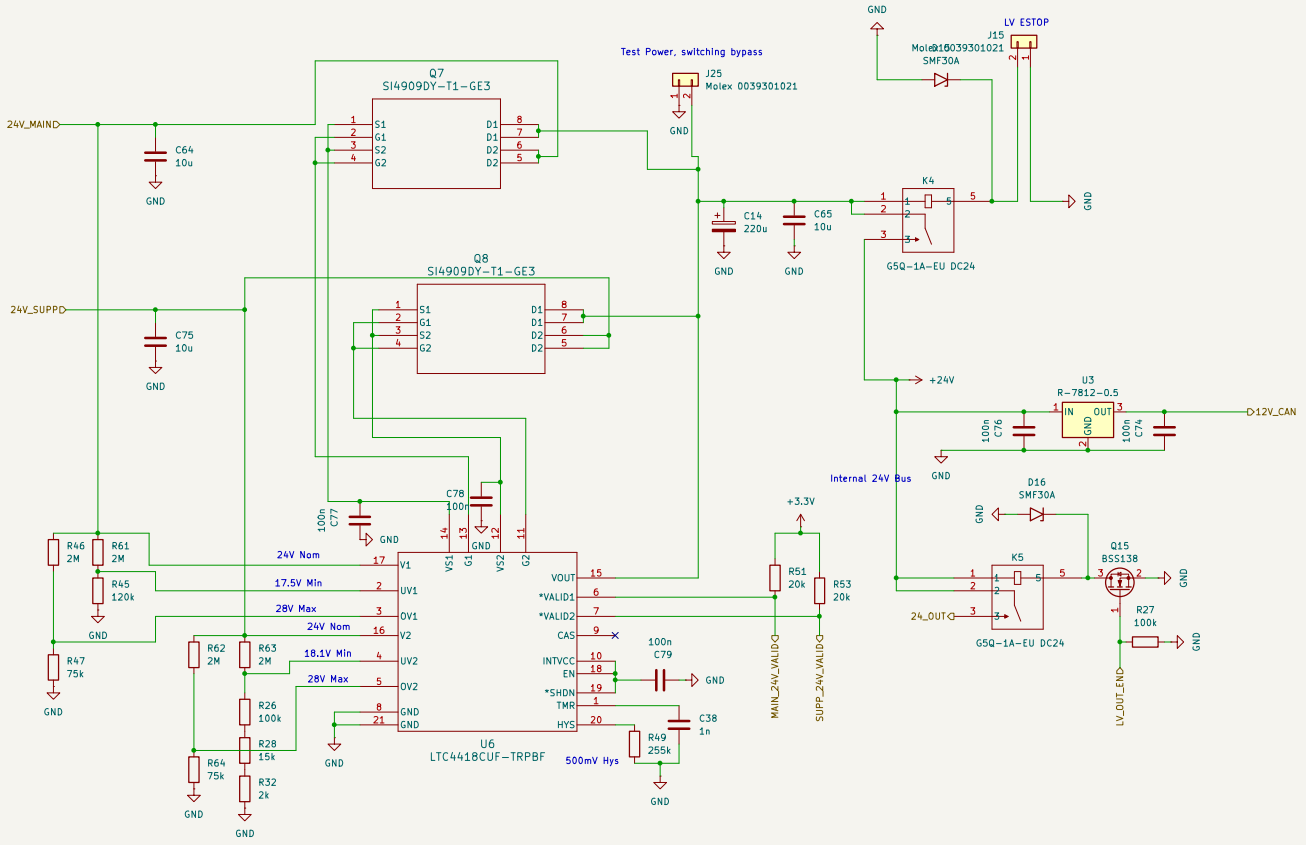


CAN Tranciever

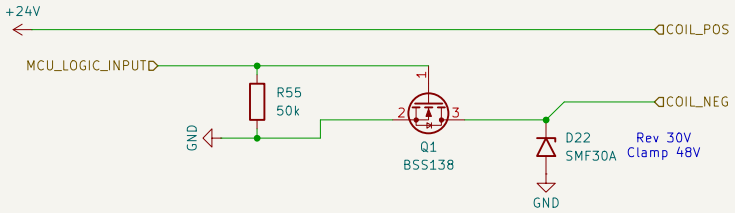
Sheet: /STM32_CONTROL/CAN Transceiver – MPPT Network/		
File: CAN Transceiver – MCP2562.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 2/27



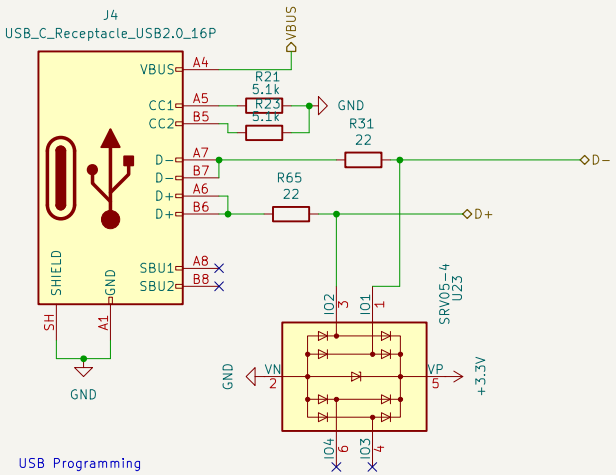
Sheet: /STM32_CONTROL/CAN Transceiver – Car Network/		
File: CAN Transceiver – MCP2562 – Non term.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 3/27



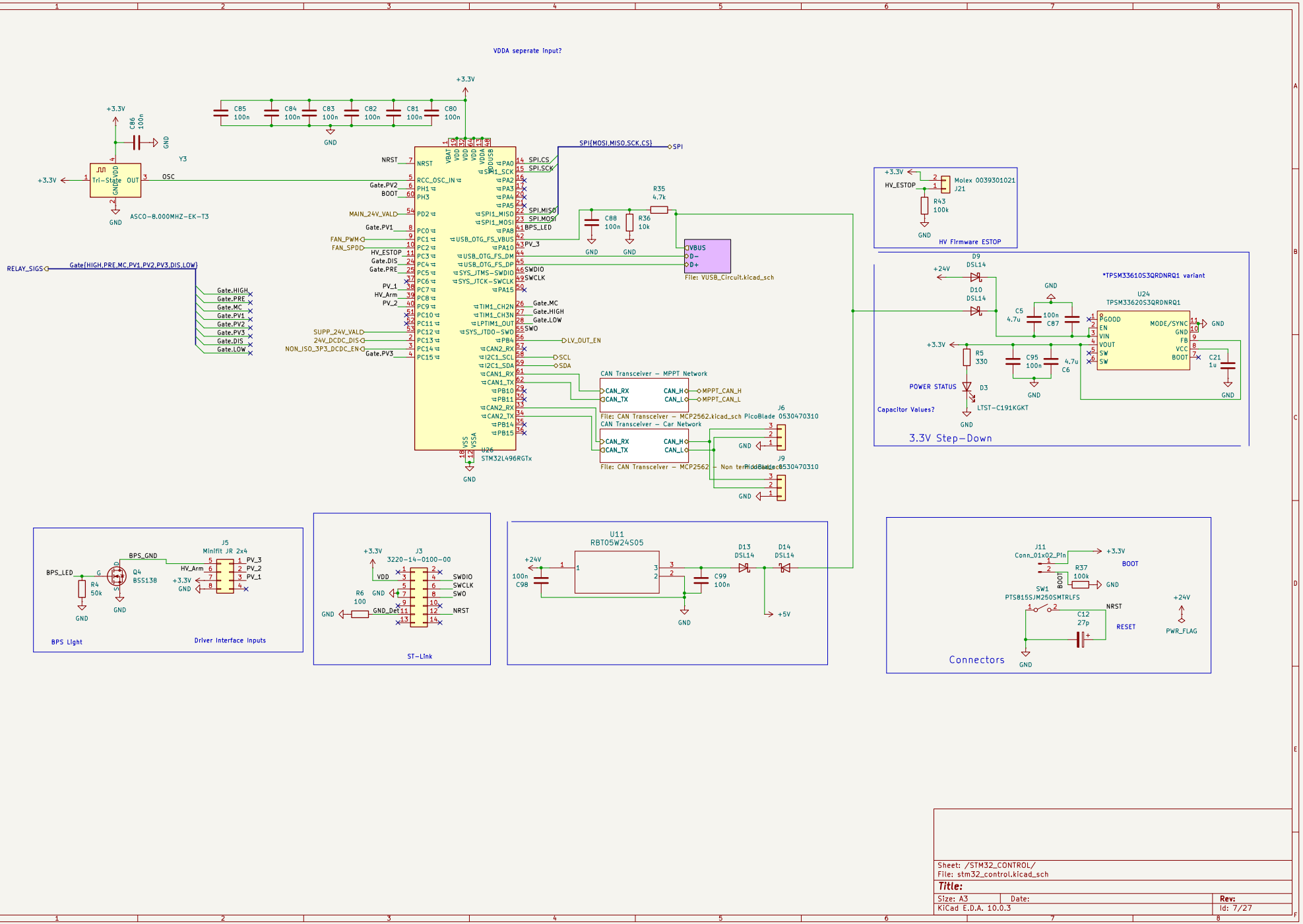
Sheet: /24 Supplementary Switch/		
File: supp.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 4/27



Sheet: /Elmar Racing MPPTS/Relay Control MPPT1/		
File: RelayProtectionCircuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 5/27

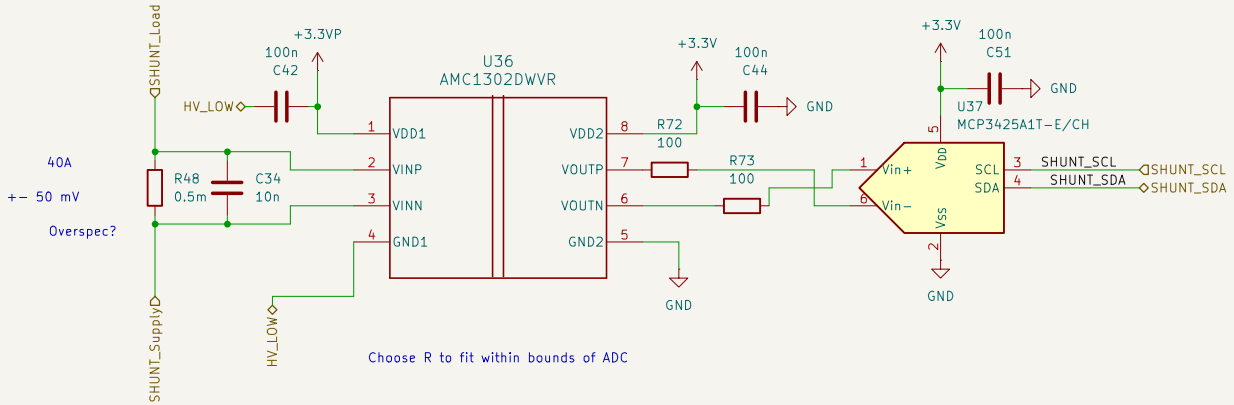


Sheet: /STM32_CONTROL/USC_CIRCUIT/		
File: VUSB_Circuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/27

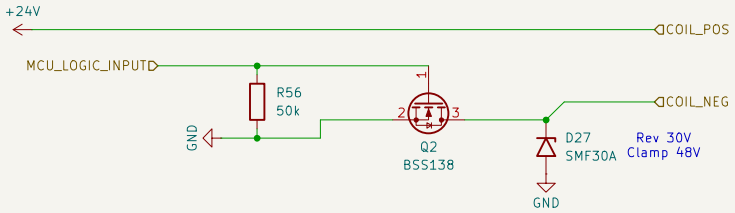


Sheet: /STM32_CONTROL/		
File: stm32_control.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 7/27

*R 10K, 100KHZ.
2K, 400KHZ.



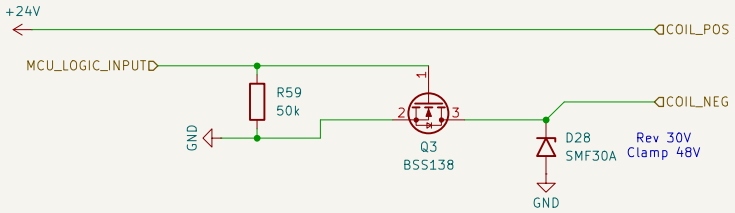
Sheet: /MC Current Sense/		
File: shunt_sensor_circuit_40A.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 8/27



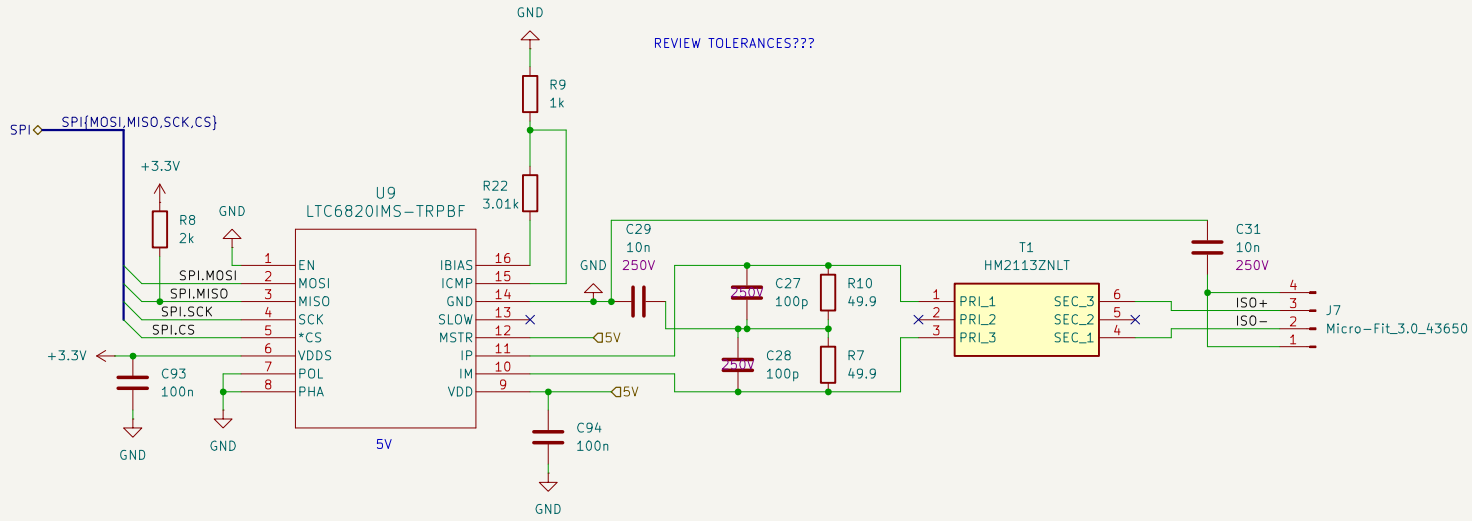
Sheet: /Elmar Racing MPPTS/Relay Control MPPT2/
File: RelayProtectionCircuit.kicad_sch

Title:

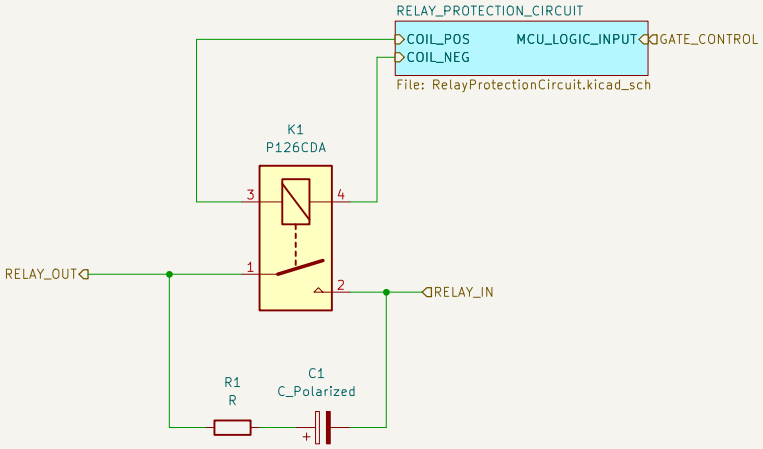
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 9/27



Sheet: /Elmar Racing MPPTS/Relay Control MPPT3/		
File: RelayProtectionCircuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 10/27

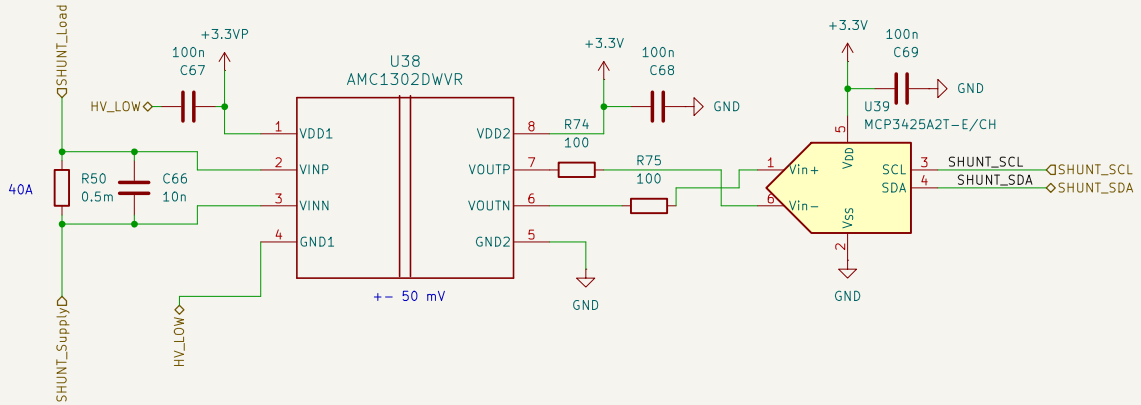


Sheet: /ISOSPI/		
File: isospi.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 11/27

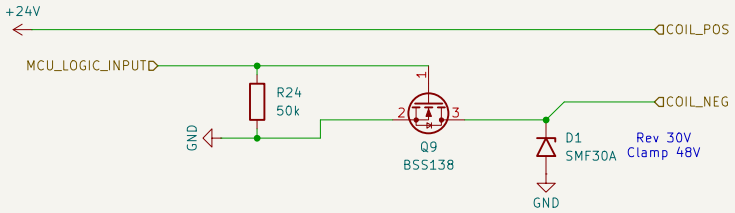


Sheet: /Main Contactor High/		
File: main_contactor.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 12/27

*R 10K, 100KHZ.
2K, 400KHZ.



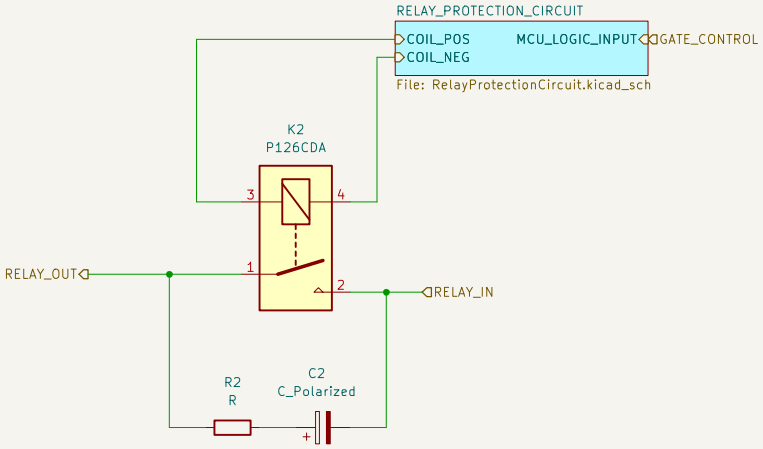
Sheet: /MPPT Current Sense/		
File: shunt_sensor_circuit_mppt.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 13/27



Sheet: /Main Contactor High/RELAY_PROTECTION_CIRCUIT/
File: RelayProtectionCircuit.kicad_sch

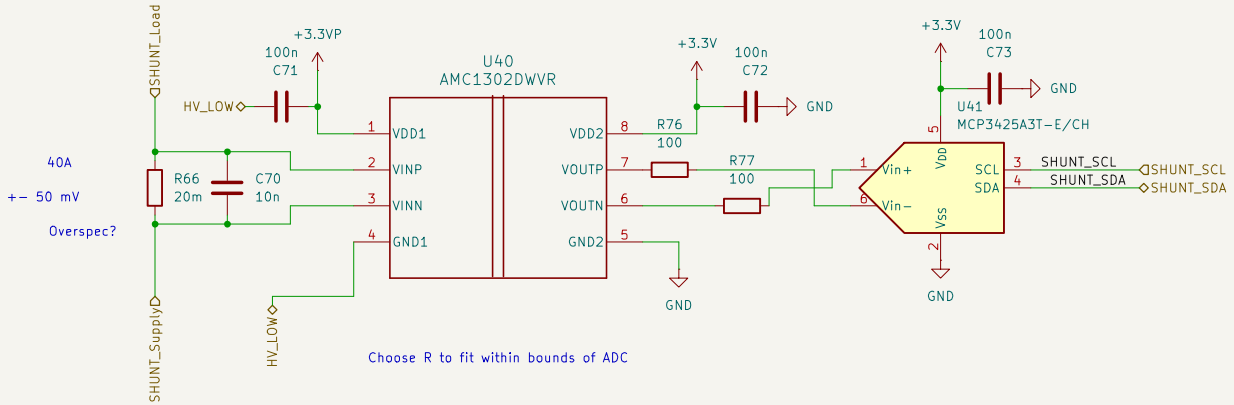
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 14/27



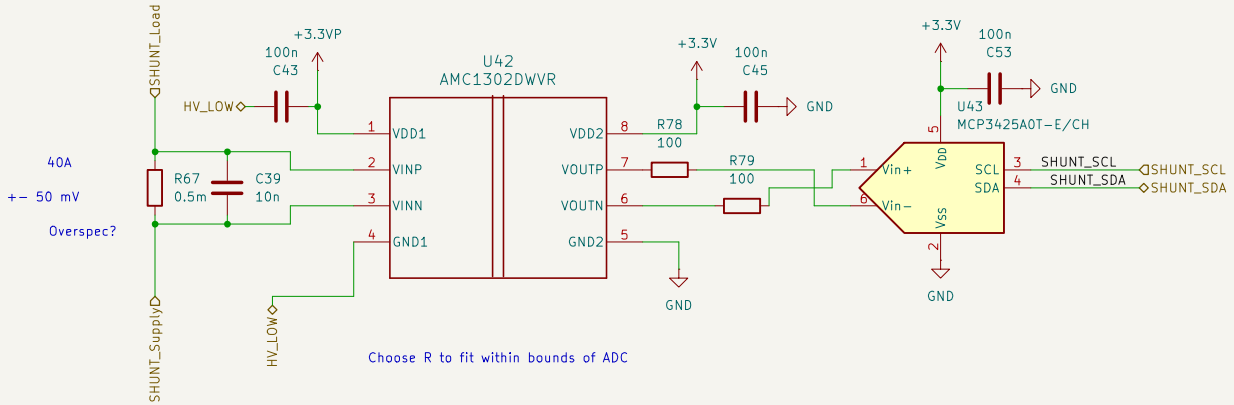
Sheet: /Main Contactor Low/		
File: main_contactor.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 15/27

*R 10K, 100KHZ.
2K, 400KHZ.

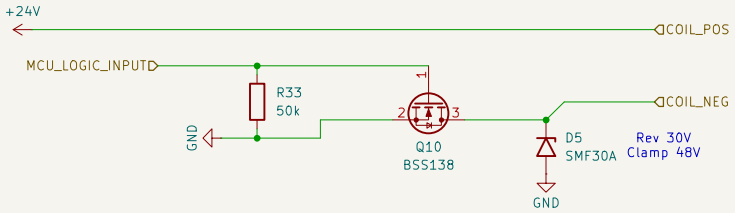


Sheet: /LV DCDC Current Sense/		
File: shunt_sensor_circuit_dcdc.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 16/27

*R 10K, 100KHZ.
2K, 400KHZ.



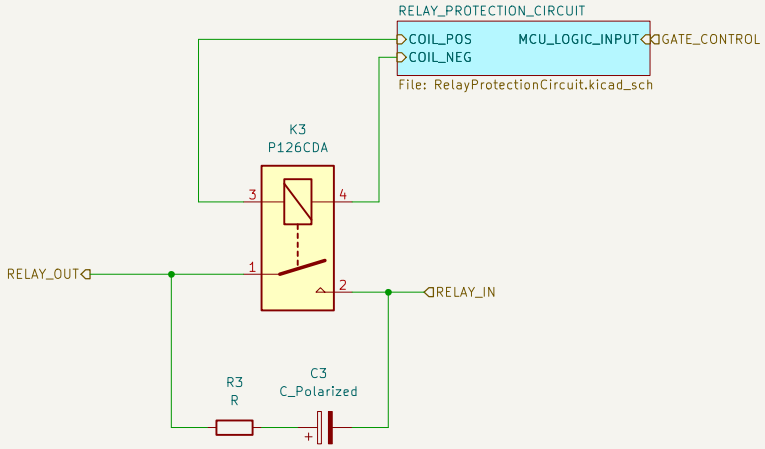
Sheet: /Main Current Sense/		
File: shunt_sensor_circuit_main.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 17/27



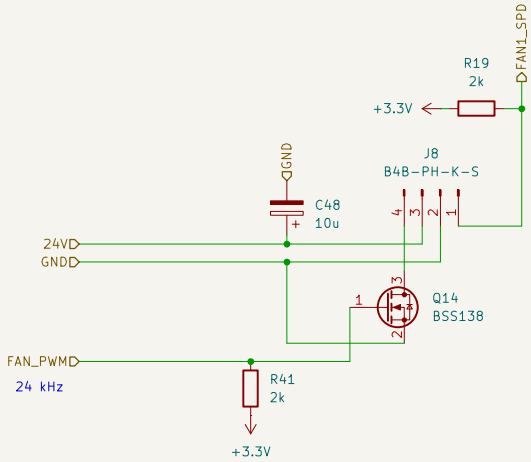
Sheet: /Main Contactor Low/RELAY_PROTECTION_CIRCUIT/
File: RelayProtectionCircuit.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 18/27



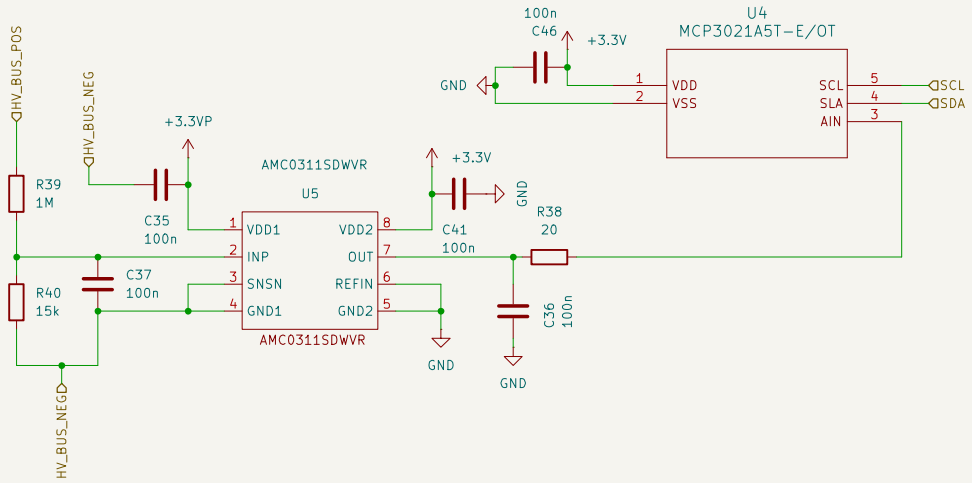
Sheet: /MC Relay/		
File: motor_controller_relay.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 19/27



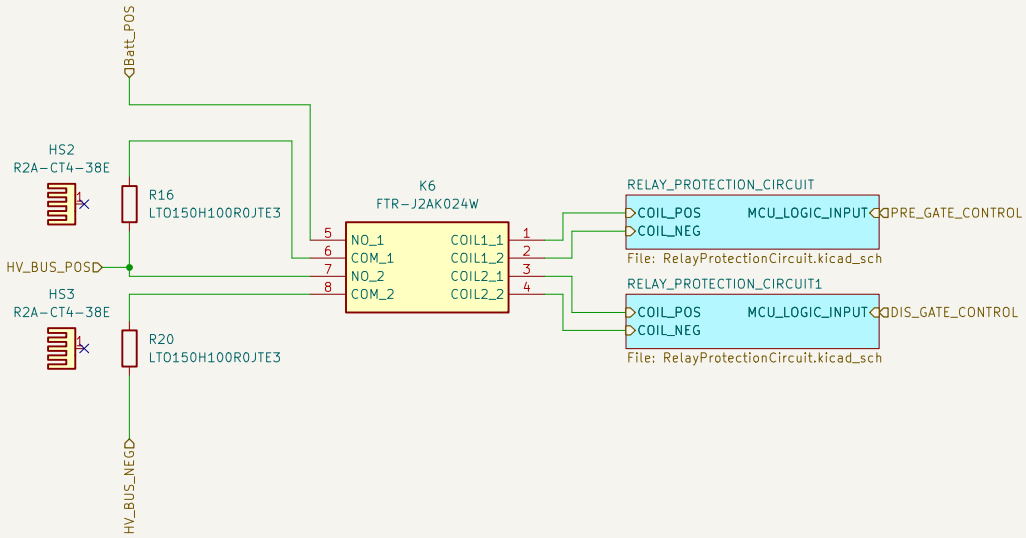
24VDC, 12W per fan

PWR, GND, SENSOR, CNTRL

Sheet: /Battery Fan/		
File: batt_fan.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 20/27



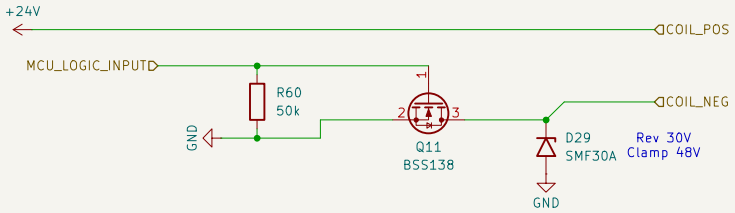
Sheet: /HV_HIGHSIDE_VOLTSENSE/		
File: HV_HIGHSIDE_VOLTSENSE.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 21/27



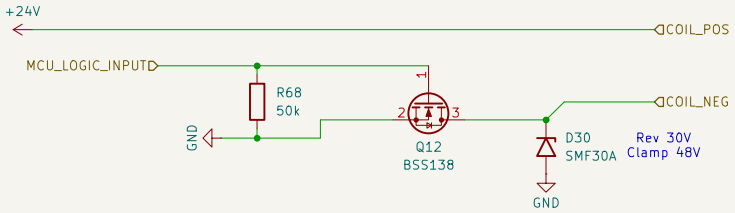
Sheet: /Pre & Dis Charge/
File: charge_circuits.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 22/27



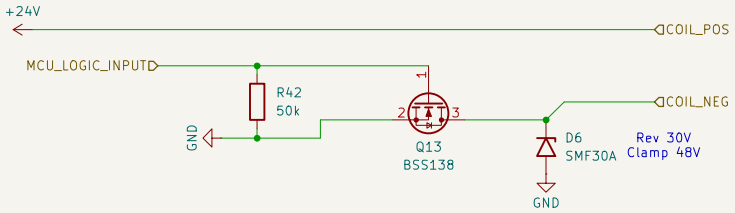
Sheet: /Pre & Dis Charge/RELAY_PROTECTION_CIRCUIT/		
File: RelayProtectionCircuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 23/27



Sheet: /Pre & Dis Charge/RELAY_PROTECTION_CIRCUIT1/
File: RelayProtectionCircuit.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 25/27



Sheet: /MC Relay/RELAY_PROTECTION_CIRCUIT/		
File: RelayProtectionCircuit.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 26/27

High Voltage Side

Low Voltage Side

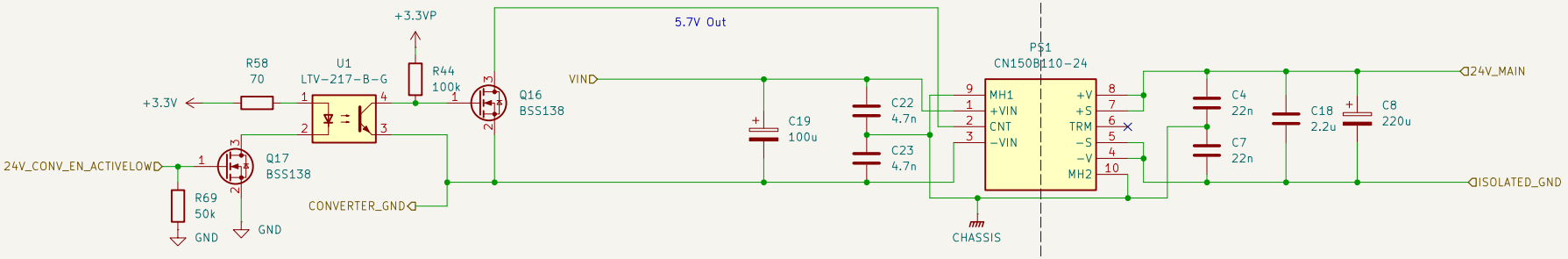
Heatsink required

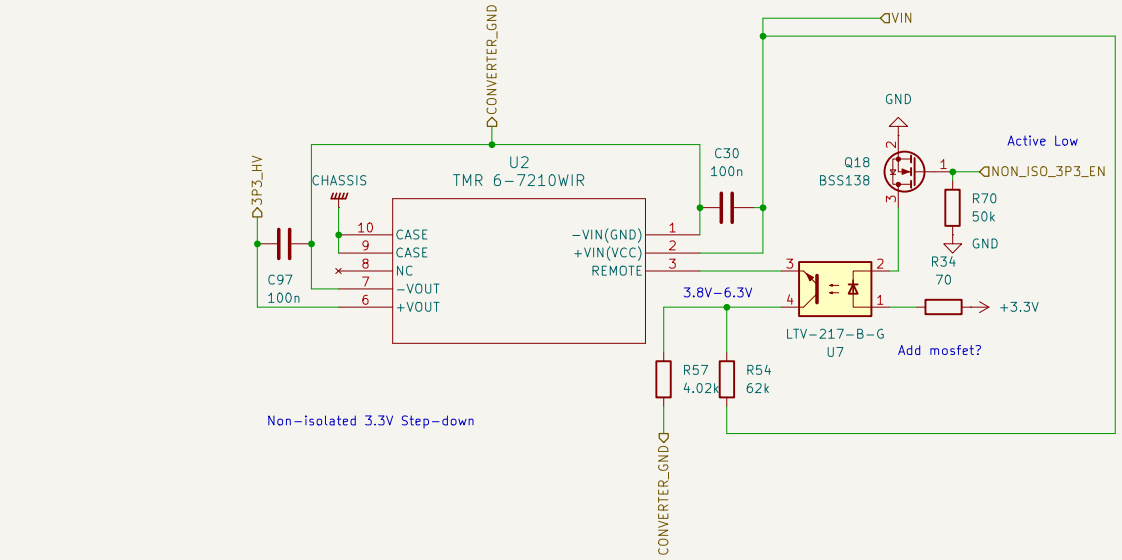
Separate Horn DCDC?

Battery negative vs chassis ground?

Bulk & bypass capacitors

POWER STATUS





Non-isolated 3.3V Step-down

24/12/3P3 Step Down

Sheet: /LV Step-down/
File: lv_step_down.kicad_sch

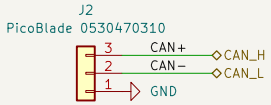
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 37/27

1	+12V	Can Supply voltage, 10.8-13.2V
2	GND	Can Ground
3	NC	Not connected
4	CAN-H	Can High
5	CAN-L	Can Low
6	NC	Not connected

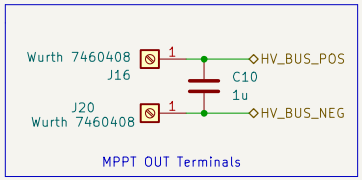
1	$V_{in}-$	Array -
2	$V_{in}+$	Array +
3	$V_{out}+$	Battery +
4	$V_{out}-$	Battery -

*Array Connections screw directly into terminal-style connector



CAN will be daisy-chained between MPPT's

What is CAN voltage?

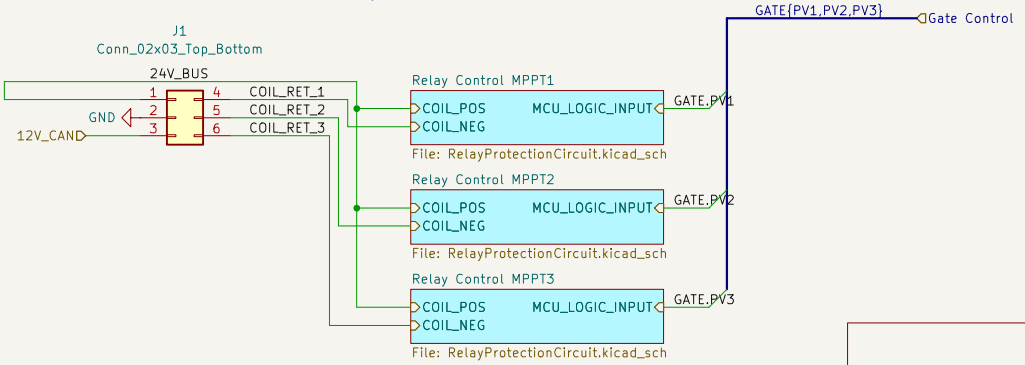


The Power connector used on the MPPT is Phoenix 1720819 (PC 5/4-GF-7,62). The recommended mating part for this connector is Phoenix contact 1777859 (PC 5/ 4-STF1-7,62). The connector is also backwards compatible with the older PC 4 counterparts.

Mating parts for this connector can be either:

- Wurth Electronics 662006113322
- Micro-Fit 3.0 430250600 / 430250608 / 430250610

Relay Control



Sheet: /Elmar Racing MPPTS/
File: elmar_racing_mppt.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 38/27